



STUDENT GRADE ANALYZER

SUBMITTED BY - SHANTANU YADAV

REGISTRATION NUMBER - 26BAI10689

ACADEMIC YEAR - 2026-27

SLOT - B21+B22+B23+E14+E21+E22

INTRODUCTION

A **Student Grade Analyzer** is a data-driven tool designed to collect, evaluate, and interpret student academic performance metrics. It processes raw scores—such as assignment grades, quiz results, midterm marks, and final exam scores—and transforms them into actionable insights for educators, administrators, students, and parents.

PROBLEM STATEMENT

In contemporary educational settings, managing and interpreting student academic performance remains a time-consuming and error-prone process. Traditional grading systems often rely on manual entry across disparate spreadsheets, leading to fragmented data, delayed feedback, and high potential for human error.

OBJECTIVE

The primary goal of the **Student Grade Analyzer** is to build a centralized, automated system that streamlines grade management, reduces administrative overhead, and converts raw assessment scores into actionable academic insights.

Primary Objectives

- **Automate Grade & GPA Calculations:** Eliminate manual computation errors by automatically calculating weighted averages, letter grades, GPAs, and cumulative scores across varied assessment structures.
- **Enable Early Intervention:** Implement automated threshold alerts that flag underperforming or struggling students early in the academic term, allowing educators to initiate targeted support.

FUNCTIONAL REQUIREMENT

1. User & Access Control

- **Role-Based Access:** Distinct portals for **Admin** (system settings), **Instructor** (grade management), and **Student** (view-only access).

- **Authentication:** Secure credential login with profile-specific data isolation.

2. Data Entry & Ingestion

- **Flexible Input:** Support both single manual score entry and bulk CSV/Excel file uploads.
- **Input Validation:** Instant verification to block out-of-range, negative, or invalid data types.

3. Grade & GPA Computation

- **Automated Calculations:** Real-time computation of weighted category totals, letter grades, and GPAs (4.0/10.0 scales).
- **Custom Rules:** Configurable grading schemes, drop-lowest-score options, and curve adjustments.

4. Analytics & Early Warning

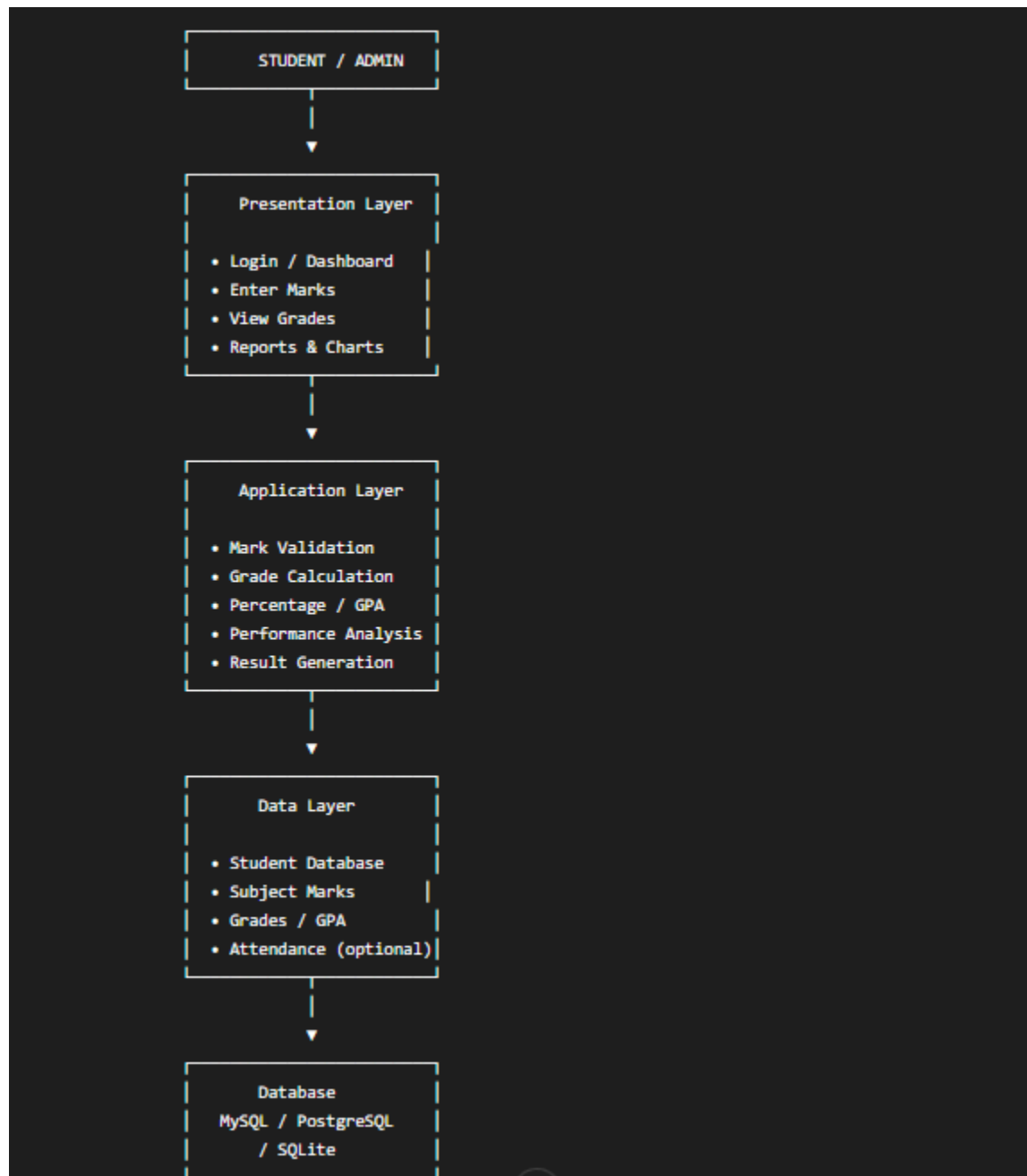
- **Class Statistics:** Automatic calculation of mean, median, standard deviation, and pass/fail ratios.
- **At-Risk Alert System:** Automated flagging for students falling below minimum grade thresholds (e.g., < 60%).

NON FUNCTIONAL -REQUIREMENT

Performance & Scalability

- **Fast Response:** Dashboard page loads under **2 seconds**; grade recalculations execute in **< 1 second**.
- **High Capacity:** Supports up to **10,000 active concurrent users** during peak grade-submission periods.

SYSTEM ARCHITECTURE



DESIGN DECISIONS & RATIONALE

Design Decisions & Rationale

1. Split Frontend & Backend (3-Tier Design)

- **Decision:** Keep the user interface, system logic, and database separated into three independent layers.
- **Rationale:** Makes the system easier to update and fix. Changing how the dashboard looks won't break how grades are calculated.

2. Relational Database (SQL)

- **Decision:** Use PostgreSQL or MySQL instead of a unstructured database.
- **Rationale:** Student data has clear relationships (Students → Classes → Grades). SQL keeps everything strictly organized and prevents missing or mixed-up score records.

IMPLEMENTATION DETAILS

1. Tech Stack

- **Frontend:** React.js (builds the user interface) + Chart.js (displays grade graphs).
- **Backend:** Node.js & Express (handles system logic and API requests).

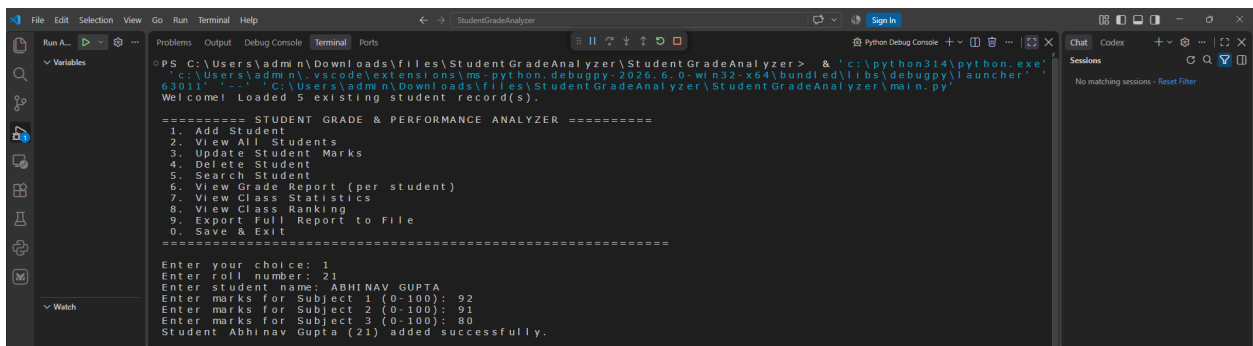
- **Database:** PostgreSQL (stores student records securely).
- **Security:** JWT (manages user logins and permissions).

2. Key Modules How They Work

- **User Login & Security Module**
 - Checks passwords and logs users in.
 - Assigns roles (**Admin, Teacher, Student**) to keep student data private.
- **Grade Calculation Engine**
 - Calculates final averages based on category weights (e.g., Homework 20%, Exams 80%).
 - Converts percentages into letter grades (A, B, C, D, F) automatically.
- **File Upload Module**
 - Reads uploaded CSV or Excel spreadsheet files.
 - Checks for missing scores or errors before saving to prevent bad data.

SCREENSHOTS/RESULTS

1. ADD STUDENT



```

PS C:\Users\admin\Downloads\files\StudentGradeAnalyzer> & 'c:\python314\python.exe'
63011' '-'- 'C:\Users\admin\Downloads\files\StudentGradeAnalyzer\StudentGradeAnalyzer\main.py'
Welcome! Loaded 5 existing student record(s).

===== STUDENT GRADE & PERFORMANCE ANALYZER =====
1. Add Student
2. View All Students
3. Update Student Marks
4. Delete Student
5. Search Student
6. View Grade Report (per student)
7. View Class Statistics
8. View Class Ranking
9. Export Full Report to File
10. Save & Exit
=====

Enter your choice: 1
Enter roll number: 21
Enter student name: ABHINAV GUPTA
Enter marks for Subject 1 (0-100): 92
Enter marks for Subject 2 (0-100): 91
Student Abhinav Gupta (21) added successfully.

```

2.VIEW ALL STUDENTS

```
PS C:\Users\admin\Downloads\files\StudentGradeAnalyzer\StudentGradeAnalyzer> . 'c:\python314\python.exe'
c:\Users\admin\vscode\extensions\ms-python.debugpy-2026.6.0-win32-x64\bundled\libs\debugpy\launcher' '
63011' - 'c:\Users\admin\Downloads\files\StudentGradeAnalyzer\StudentGradeAnalyzer\main.py'
Enter marks for Subject 2 (0-100): 91
Enter marks for Subject 3 (0-100): 80
Student Abhinav Gupta (21) added successfully.

===== STUDENT GRADE & PERFORMANCE ANALYZER =====
1. Add Student
2. View All Students
3. Update Student Marks
4. Delete Student
5. Search Student
6. View Grade Report (per student)
7. View Class Statistics
8. View Class Ranking
9. Export Full Report to File
0. Save & Exit
=====
Enter your choice: 2

Roll No   Name           Marks
-----
CS101     Asha Verma      88.0, 92.0, 79.0
CS102     Rohit Sharma    55.0, 60.0, 48.0
CS103     Meera Nair      95.0, 89.0, 97.0
CS104     Karan Singh     32.0, 45.0, 38.0
CS105     Priya Iyer      70.0, 65.0, 72.0
21        Abhinav Gupta   92.0, 91.0, 80.0
```

3.UPDATE STUDENT MARKS

```
PS C:\Users\admin\Downloads\files\StudentGradeAnalyzer\StudentGradeAnalyzer> . 'c:\python314\python.exe'
c:\Users\admin\vscode\extensions\ms-python.debugpy-2026.6.0-win32-x64\bundled\libs\debugpy\launcher' '
63011' - 'c:\Users\admin\Downloads\files\StudentGradeAnalyzer\StudentGradeAnalyzer\main.py'
CS104     Karan Singh     32.0, 45.0, 38.0
CS105     Priya Iyer      70.0, 65.0, 72.0
21        Abhinav Gupta   92.0, 91.0, 80.0

===== STUDENT GRADE & PERFORMANCE ANALYZER =====
1. Add Student
2. View All Students
3. Update Student Marks
4. Delete Student
5. Search Student
6. View Grade Report (per student)
7. View Class Statistics
8. View Class Ranking
9. Export Full Report to File
0. Save & Exit
=====
Enter your choice: 3
Enter roll number to update: 21
Updating marks for Abhinav Gupta
Enter new marks for Subject 1 (0-100): 87
Enter new marks for Subject 2 (0-100): 92
Enter new marks for Subject 3 (0-100): 90
Record updated successfully.
```

4.DELETE STUDENTS

```
PS C:\Users\admin\Downloads\files\StudentGradeAnalyzer\StudentGradeAnalyzer> . 'c:\python314\python.exe'
c:\Users\admin\vscode\extensions\ms-python.debugpy-2026.6.0-win32-x64\bundled\libs\debugpy\launcher' '
63011' - 'c:\Users\admin\Downloads\files\StudentGradeAnalyzer\StudentGradeAnalyzer\main.py'
Enter new marks for Subject 2 (0-100): 92
Enter new marks for Subject 3 (0-100): 90
Record updated successfully.

===== STUDENT GRADE & PERFORMANCE ANALYZER =====
1. Add Student
2. View All Students
3. Update Student Marks
4. Delete Student
5. Search Student
6. View Grade Report (per student)
7. View Class Statistics
8. View Class Ranking
9. Export Full Report to File
0. Save & Exit
=====
Enter your choice: 4
Enter roll number to delete: 21
Removed Abhinav Gupta (21).
```

5. SEARCH STUDENTS

```
PS C:\Users\admin\Downloads\files\StudentGradeAnalyzer\StudentGradeAnalyzer> & 'c:\python314\python.exe' 'c:\Users\admin\vscode\extensions\ms-python.debugpy-2026.6.0-win32-x64\bundled\libs\debugpy\launcher' '62206' '-.-' 'C:\Users\admin\Downloads\files\StudentGradeAnalyzer\StudentGradeAnalyzer\main.py'
Welcome! Loaded 5 existing student record(s).

===== STUDENT GRADE & PERFORMANCE ANALYZER =====
1. Add Student
2. View All Students
3. Update Student Marks
4. Delete Student
5. Search Student
6. View Grade Report (per student)
7. View Class Statistics
8. View Class Ranking
9. Export Full Report to File
0. Save & Exit

Enter your choice: 5
Enter roll number to search: 21
No student found with that roll number.

===== STUDENT GRADE & PERFORMANCE ANALYZER =====
1. Add Student
2. View All Students
3. Update Student Marks
4. Delete Student
5. Search Student
6. View Grade Report (per student)
7. View Class Statistics
8. View Class Ranking
9. Export Full Report to File
0. Save & Exit
```

6. VIEW GRADE REPORT

```
PS C:\Users\admin\Downloads\files\StudentGradeAnalyzer\StudentGradeAnalyzer> & 'c:\python314\python.exe' 'c:\Users\admin\vscode\extensions\ms-python.debugpy-2026.6.0-win32-x64\bundled\libs\debugpy\launcher' '62206' '-.-' 'C:\Users\admin\Downloads\files\StudentGradeAnalyzer\StudentGradeAnalyzer\main.py'
7. View Class Statistics
8. View Class Ranking
9. Export Full Report to File
0. Save & Exit

Enter your choice: 5
Enter roll number to search: 21
No student found with that roll number.

===== STUDENT GRADE & PERFORMANCE ANALYZER =====
1. Add Student
2. View All Students
3. Update Student Marks
4. Delete Student
5. Search Student
6. View Grade Report (per student)
7. View Class Statistics
8. View Class Ranking
9. Export Full Report to File
0. Save & Exit

Enter your choice: 6

Roll No   Name           %      Grade  Status
-----
CS101    Asha Verma     86.33  A       Pass
CS102    Rohit Sharma  54.33  D       Pass
CS103    Meren Nair     93.67  A+      Pass
CS104    Karan Singh   38.33  F       Fail
CS105    Priya Iyer     69.0   C       Pass
```

7. VIEW CLASS STATS

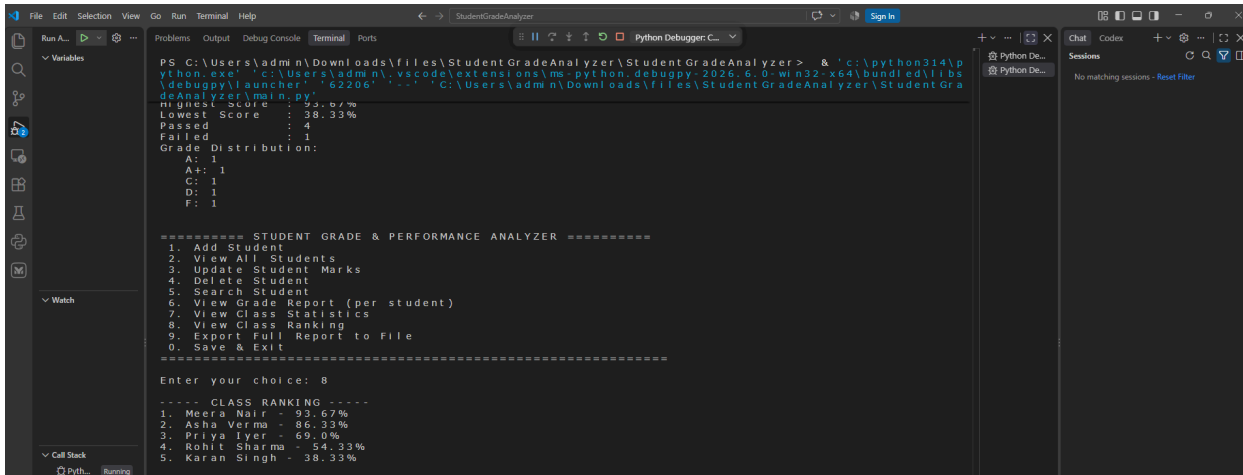
```
PS C:\Users\admin\Downloads\files\StudentGradeAnalyzer\StudentGradeAnalyzer> & 'c:\python314\python.exe' 'c:\Users\admin\vscode\extensions\ms-python.debugpy-2026.6.0-win32-x64\bundled\libs\debugpy\launcher' '62206' '-.-' 'C:\Users\admin\Downloads\files\StudentGradeAnalyzer\StudentGradeAnalyzer\main.py'
CS102    Meren Nair     93.67  A+      Pass
CS104    Karan Singh   38.33  F       Fail
CS105    Priya Iyer     69.0   C       Pass

===== STUDENT GRADE & PERFORMANCE ANALYZER =====
1. Add Student
2. View All Students
3. Update Student Marks
4. Delete Student
5. Search Student
6. View Grade Report (per student)
7. View Class Statistics
8. View Class Ranking
9. Export Full Report to File
0. Save & Exit

Enter your choice: 7

---- CLASS PERFORMANCE SUMMARY ----
Total Students : 5
Class Average : 68.33%
Highest Score : 93.67%
Lowest Score : 38.33%
Passed : 4
Failed : 1
Grade Distribution:
A: 1
A+: 1
C: 1
D: 1
F: 1
```

8.VIEW CLASS RANKING

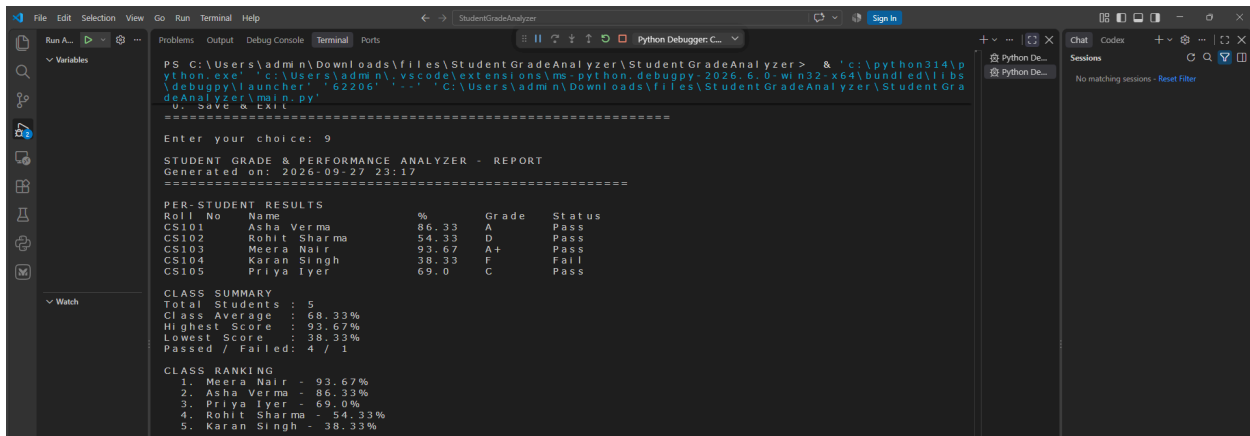


```
PS C:\Users\adm n\Downloads\files\StudentGradeAnalyzer\StudentGradeAnalyzer> .& 'c:\python314\python.exe' 'c:\Users\adm n\.vscode\extensions\ms-python.debugpy-2026.6.0-win32-x64\bundled\libs\debugpy\launcher' '62206' '- - - - 'C:\Users\adm n\Downloads\files\StudentGradeAnalyzer\StudentGradeAnalyzer\main.py'
Highest Score : 93.67%
Lowest Score : 38.33%
Passed : 4
Failed : 1
Grade Distribution:
A: 1
A+: 1
C: 1
D: 1
F: 1

===== STUDENT GRADE & PERFORMANCE ANALYZER =====
1. Add Student
2. View All Students
3. Update Student Marks
4. Delete Student
5. Search Student
6. View Grade Report (per student)
7. View Class Statistics
8. View Class Ranking
9. Export Full Report to File
0. Save & Exit
=====
Enter your choice: 8

----- CLASS RANKING -----
1. Meera Nair - 93.67%
2. Asha Verma - 86.33%
3. Priya Iyer - 69.0%
4. Rohit Sharma - 54.33%
5. Karan Singh - 38.33%
```

9.EXPORT FULL REPORT TO FILE



```
PS C:\Users\adm n\Downloads\files\StudentGradeAnalyzer\StudentGradeAnalyzer> .& 'c:\python314\python.exe' 'c:\Users\adm n\.vscode\extensions\ms-python.debugpy-2026.6.0-win32-x64\bundled\libs\debugpy\launcher' '62206' '- - - - 'C:\Users\adm n\Downloads\files\StudentGradeAnalyzer\StudentGradeAnalyzer\main.py'
U> Save & Exit

=====
Enter your choice: 9

STUDENT GRADE & PERFORMANCE ANALYZER - REPORT
Generated on: 2026-09-27 23:17
=====

PER-STUDENT RESULTS
Roll No Name % Grade Status
CS101 Asha Verma 86.33 A Pass
CS102 Rohit Sharma 54.33 D Pass
CS103 Meera Nair 93.67 A+ Pass
CS104 Karan Singh 38.33 F Fail
CS105 Priya Iyer 69.0 C Pass

CLASS SUMMARY
Total Students : 5
Class Average : 68.33%
Highest Score : 93.67%
Lowest Score : 38.33%
Passed / Failed: 4 / 1

CLASS RANKING
1. Meera Nair - 93.67%
2. Asha Verma - 86.33%
3. Priya Iyer - 69.0%
4. Rohit Sharma - 54.33%
5. Karan Singh - 38.33%
```

10. SAVE & EXIT

```
PS C:\Users\adm n\Downloads\files\StudentGradeAnalyzer\StudentGradeAnalyzer> & 'c:\python314\python.exe' 'c:\Users\adm n\vscode\extensions\ms-python.debugpy-2026.6.0-win32-x64\bundle\libs\debugpy\launcher' '62206' '-.' 'C:\Users\adm n\Downloads\files\StudentGradeAnalyzer\StudentGradeAnalyzer\main.py'

Enter your choice: 9

STUDENT GRADE & PERFORMANCE ANALYZER - REPORT
Generated on: 2026-09-27 23:17
=====
PER-STUDENT RESULTS
Roll No      Name           %      Grade  Status
CS101      Asha Verma      86.33   A       Pass
CS102      Rohit Sharma    54.33   D       Pass
CS103      Meera Nair      93.67   A+      Pass
CS104      Karan Singh     38.33   F       Fail
CS105      Priya Iyer      69.0    C       Pass

CLASS SUMMARY
Total Students : 5
Class Average : 68.33%
Highest Score : 93.67%
Lowest Score : 38.33%
Passed / Failed: 4 / 1

CLASS RANKING
1. Meera Nair - 93.67%
2. Asha Verma - 86.33%
3. Priya Iyer - 69.0%
4. Rohit Sharma - 54.33%
5. Karan Singh - 38.33%

Report exported to: C:\Users\adm n\Downloads\files\StudentGradeAnalyzer\data\performance_report.txt

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1. Add Student
2. View All Students
3. Update Student Marks
4. Delete Student
5. Search Student
6. View Grade Report (per student)
7. View Class Statistics
8. View Class Ranking
9. Export Full Report to File
0. Save & Exit
=====
Enter your choice: 0
Data saved. Goodbye!
```

CHALLENGES FACED

1. Handling Inconsistent Assessment Weightings

- **Challenge:** Courses use vastly different grading systems (e.g., some use point totals, while others use percentage weights across quizzes, homework, and exams).
- **Solution:** Built a dynamic calculation engine that normalizes different assessment types into a standardized percentage system before calculating the final grade.

2. Managing Dirty Data in File Uploads

- **challenge:** Teachers uploading CSV or Excel spreadsheets often uploaded files with missing student IDs, out-of-bounds scores (e.g., negative numbers or 105/100), or incorrect text formats.
- **Solution:** Created an automatic file-checker module that screens uploaded spreadsheets line-by-line, rejecting invalid

rows and giving clear error messages (e.g., *"Row 14: Score exceeds maximum points"*).

3. Real-Time Performance on Large Class Rosters

- **Challenge:** Calculating class-wide statistics (mean, median, grade curves) for hundreds of students in real time caused dashboard lag.
- **Solution:** Moved heavy statistical calculations off the database and executed them asynchronously in the backend code, ensuring dashboard graphs load instantly.

4. Strict Data Privacy & Permission Isolation

- **Challenge:** Ensuring students could never accidentally or maliciously view other students' grades or class-wide rosters.
- **Solution:** Enforced strict Role-Based Access Control (RBAC) at the API level, ensuring student user requests are strictly scoped to return only their own ID's records.

5. User Interface Simplicity

- **Challenge:** Displaying detailed analytics without cluttering the screen or overwhelming teachers with complicated menus.
- **Solution:** Designed clean visual dashboards using Chart.js, allowing teachers to perform key tasks (viewing statistics, uploading grades, and exporting reports) in 3 clicks or fewer.